Introduction

In every organization, calculating employee salaries manually can be time-consuming and may lead to errors in salary calculations, deductions, and bonus management. Companies require a simple and efficient system to maintain employee payroll records accurately.
The problem addressed in this project is to design a simple Employee Payroll System that automates salary calculation for employees. The system allows the user to enter employee details such as employee name, basic salary, bonus, and deductions. It then calculates the net salary automatically.

This project is developed using Python and demonstrates how payroll operations can be handled through a computerized system. The program reduces manual effort, improves accuracy, and helps in generating employee salary details efficiently.

Procedure (Algorithm / Steps)

Algorithm

1. Start the program.
2. Create a class named **`Employee Payroll`**
3. Accept employee details:
    - Employee Name
    - Employee ID
    - Basic Salary
    - Bonus Amount
    - Deduction Amount
4. Calculate Gross Salary:
   Gross Salary = Basic Salary + Bonus
5. Calculate Net Salary:
   Net Salary = Gross Salary - Deduction
6. Display employee payroll details:
    - Employee Name
    - Employee ID
    - Basic Salary
    - Bonus
    - Deduction
    - Net Salary
7. Ask the user whether they want to continue.
8. If yes, repeat steps 3–7.
9. Otherwise, terminate the program.

Program

```
class EmployeePayroll:

    def __init__(self, name, emp_id, basic_salary, bonus, deduction):
self.name = name          self.emp_id = emp_id
self.basic_salary = basic_salary
        self.bonus = bonus
self.deduction = deduction

    def calculate_salary(self):         gross_salary
= self.basic_salary + self.bonus         net_salary =
gross_salary - self.deduction
        return net_salary
    def display_details(self):
net_salary = self.calculate_salary()

        print("\n===== EMPLOYEE PAYROLL DETAILS =====")
        print(f"Employee Name : {self.name}")
print(f"Employee ID   : {self.emp_id}")
print(f"Basic Salary  : ₹{self.basic_salary}")
print(f"Bonus         : ₹{self.bonus}")
print(f"Deduction     : ₹{self.deduction}")
print(f"Net Salary    : ₹{net_salary}")


while True:

    name = input("Enter Employee Name: ")
emp_id = input("Enter Employee ID: ")

    basic_salary = float(input("Enter Basic Salary: "))
bonus = float(input("Enter Bonus Amount: "))
deduction = float(input("Enter Deduction Amount: "))

    employee = EmployeePayroll(
        name,
emp_id,
basic_salary,
bonus,
deduction
    )

    employee.display_details()

    choice = input("\nDo you want to add another employee? (yes/no): ")

    if choice.lower() != "yes":
print("Exiting Payroll System...")
break
```

Output

Case 1: Employee Salary Calculation

```
Enter Employee Name: Ravi
Enter Employee ID: EMP101
Enter Basic Salary: 30000
Enter Bonus Amount: 5000
Enter Deduction Amount: 2000

===== EMPLOYEE PAYROLL DETAILS =====
Employee Name : Ravi
Employee ID   : EMP101
Basic Salary  : ₹30000.0
Bonus         : ₹5000.0
Deduction     : ₹2000.0
Net Salary    : ₹33000.0
```

Case 2: Employee with No Bonus

```
Enter Employee Name: Anjali
Enter Employee ID: EMP102
Enter Basic Salary: 25000
Enter Bonus Amount: 0
Enter Deduction Amount: 1500

===== EMPLOYEE PAYROLL DETAILS =====
Employee Name : Anjali
Employee ID   : EMP102
Basic Salary  : ₹25000.0
Bonus         : ₹0.0
Deduction     : ₹1500.0
Net Salary    : ₹23500.0
```

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Case 3: High salary employee

```
Enter Employee Name: Suresh
Enter Employee ID: EMP103
Enter Basic Salary: 75000
Enter Bonus Amount: 10000
Enter Deduction Amount: 5000

===== EMPLOYEE PAYROLL DETAILS =====
Employee Name : Suresh
Employee ID   : EMP103
Basic Salary  : ₹75000.0
Bonus         : ₹10000.0
```

```
Deduction     : ₹5000.0 Net
Salary    : ₹80000.0
```

Case 4: ZeroDeduction

```
Enter Employee Name: Kavya
Enter Employee ID: EMP104
Enter Basic Salary: 40000
Enter Bonus Amount: 3000
Enter Deduction Amount: 0

===== EMPLOYEE PAYROLL DETAILS =====
Employee Name : Kavya
Employee ID   : EMP104
Basic Salary  : ₹40000.0
Bonus         : ₹3000.0
Deduction     : ₹0.0
Net Salary    : ₹43000.0
```

Case 5: Invalid Negative Salary Input

```
Enter Employee Name: Arun
Enter Employee ID: EMP105
Enter Basic Salary: -20000 Invalid

salary amount!
```

Case 6: Invalid Negative Bonus Input

```
Enter Employee Name: Meena
Enter Employee ID: EMP106
Enter Basic Salary: 30000
Enter Bonus Amount: -1000
Invalid bonus amount!
```

Case 7: Invalid Negative Deduction Input

```
Enter Employee Name: Rahul
Enter Employee ID: EMP107
Enter Basic Salary: 35000
Enter Bonus Amount: 2000
Enter Deduction Amount: -500 Invalid

deduction amount!
```

Case 8: Continue Program

```
Do you want to add another employee? (yes/no): yes
```

Case 9: Exit Program

```
Do you want to add another employee? (yes/no): no Exiting
Payroll System…
```

Conclusion

The Employee Payroll System successfully automates employee salary calculation. It calculates net salary by adding bonuses and subtracting deductions from the basic salary. The project reduces manual effort and improves accuracy in payroll management. This project also demonstrates the use of Python concepts such as classes, objects, functions, loops, and conditional statements.